MILITARY SPECIFICATION

TEST SET, ATOMIC WEAPON: T-4186

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of test set entitled Test Set, Atomic Weapon T-4186 which is used with the Lance warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For.
MIL-C-48526	- Cable Assembly, Special Purpose, Electrical: W30U Through W33U, W36U, W38U, W45U, W46U, W58U, W70U, W71U and W73U.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

8877483	- Test Set, Atomic Weapon T-4186.
8877492	- Panel, Test Set.
9210852	- Case, Test Set.
9210985	- Plate, Identification.

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9221171 - Box, Packing Ammunition for Test Set, Atomic Weapon: T-4186.
EL-8877483 - Equipment List.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The test set shall comply with all requirements specified on the drawing (dwg) 8877483, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 Continuity and resistance.

3.5.1 Connector J1. Continuity between terminals connected in the combinations listed in Table I is indicated if the green "GO" lens illuminates. The test shall be conducted as specified in 4.4.1.1.

Table I

Switch Position	Between Terminal	And Terminal
2	L	M
3	L	A
4	A	B
5	B	C
6	C	E
7	E	F
8	F	G
9	G	J
10	J	P
11	P	V
12	V	U
13	U	R

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14	R	b
15	b	T
16	T	M
17	S	M
18	K	S

3.5.2 Dummy Connector Recp. (J2). The resistance between terminals listed in Table II shall be 0.025 ohm maximum when tested as specified in 4.4.1.2.

Table II

Between Terminal	And Terminal
A	B
B	C
C	E
E	F
F	G
G	J
J	K
K	L
L	M
M	P
P	R
R	S
S	T
T	U
U	V
V	b
b	C

3.5.3 Dummy Connector (J3). The resistance between terminals listed in Table III shall be 0.025 ohms maximum when tested as specified in 4.4.1.2.

Table III

Between Terminals	And Terminals
30	31
31	33
33	34
34	36
36	37
37	38
38	40
40	42
42	44
44	45
45	47
47	49

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49	52
52	53
53	54
54	55

3.5.4 Connector Shorting (J4). The resistance between terminals listed in Table IV shall be 0.025 ohms maximum when tested as specified in 4.4.1.2.

Table IV

Between Terminals	And Terminals
L	K

3.6 Insulation resistance. The resistance when measured between each of the used terminals and between all other unused terminals and the shells shall be 100 megohms minimum when tested as specified in 4.4.2. The terminals of the connectors are as listed.

Connector (J1)

Used terminals: A,B,C,E,F,G,J,L,M,P,R,T,U,V,b,S and K.
Unused terminals: D,H,N,W,X,Y,Z and a, c.

Dummy Connector (J3)

Used terminals: 30,31,33,34,36,37,38,40,42,44,45,47,49,52,53,54,55
Unused terminals: 1 thru 29 incl; 32,35,39,41,43,46,48,50,51 and 56 thru 67 incl.

Unused terminals shall be connected together and tested as one terminal.

Connector Shorting (J4)

Used terminals: L and K
Unused terminals: All except L and K

Unused terminals shall be connected together, and tested as one terminal.

Dummy Connector (J2)

Between used Pin A and all unused pins D,H,N,W,X,Y,Z,a connected together and tested as one terminal.

3.7 Components. Cable Assemblies W70U and W71U, prior to being utilized with the T-4186 test set shall have complied with all their requirements as specified in MIL-C-48526.

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3.8 Environmental testing. The test set shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The test set shall also comply with 4.3.3.1 before and after each environmental test. Testing shall be as specified in 4.4.3. Tests shall be conducted in the order specified.

3.8.1 Impact shock. The test set shall withstand 18 shocks of 15 gravity units (g) ± 1.5g acceleration. Each shock shall last 11 milliseconds (ms) ± 1 ms in duration as measured between points which is 10 percent of the peak level. The time to peak shall be reached within 5.5 ms ± 1 ms. Each shock shall be applied along the three mutually perpendicular axis, three times in each direction. After this test is completed as specified, the test set shall comply with the requirements of 3.5 and 3.6.

3.8.2 Transportation vibration. The test set shall withstand vibrations in each of the three mutually perpendicular axes to the vibration spectrum specified in Table V.

Table V

Frequency (Hertz)	Cycle Time (Minutes)	Constant Acceleration or Double Amplitude
8-55-8	7.5 ± 0.5	0.060 ± 0.006 inch
55-500-55	7.5 ± 0.5	5.0g ± 0.5g

After this test is conducted as specified, the test set shall comply with the requirements of 3.5 and 3.6.

3.8.3 High temperature. The test set shall withstand high temperature of +160 degrees (°) Fahrenheit (F) ± 5°F. After this test is conducted as specified, the test shall comply with the requirements of 3.5 and 3.6.

3.8.4 Low temperature. The test set shall withstand low temperature of -65 ± 5°F. After this test is conducted as specified, the test set shall comply with the requirements of 3.5 and 3.6.

3.8.5 Humidity. The test set shall withstand the relative humidity of 95 percent at a temperature of +160°F ± 5°F. After this test is conducted as specified in 4.4.3.5, the test set shall comply with requirements of 3.5 and 3.6.

3.9 Soldering. Soldering shall be performed in accordance with MIL-STD-1460.

3.10 Workmanship.

3.10.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.10.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

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3.10.3 Burrs. Parts shall be free from burrs.

3.10.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.10.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.10.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 8877483) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table VI. Tests shall be conducted in order specified in Table VI.

Table VI

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Continuity and resistance (see 3.5)		X
3. Insulation resistance (see 3.6)		X

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4.	Impact shock (see 3.8.1)	X
5.	Transportation vibration (see 3.8.2)	X
6.	High temperature (see 3.8.3)	X
7.	Low temperature (see 3.8.4)	X
8.	Humidity (see 3.8.5)	X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation. The formation of an inspection lot and related functions shall be in accordance with Section 5, MIL-STD-105. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table VII and Table VIII), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table VII
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table VIII
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Test set, atomic weapon T-4186 (see dwg 8877483).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Any component, missing, loose, damaged or improper	Visual
102.	Any component improperly located or assembled	Visual
103.	Markings improper, illegible and missing (see dwg 9210852 and 8877492)	Visual
104.	Improper identification (see dwg 9210985)	Visual
105.	Improper soldering (see 3.9)	Visual
106.	Incorrect torque (18 places)	Gage
107.	Latches engage and disengage readily without any undue effort	Manual
108.	Objective evidence relative to material missing or improper (see 3.1)	Visual
109.	Serialization improper or missing (see 3.4)	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.10)	Visual

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4.3.2.1.2 Preparation for delivery (see dwg 9221171).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Packing and packaging in accordance with dwg 9221171	Visual
	102. Marking in accordance with dwg 9221171	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.10)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Continuity and resistance	Major
b. Insulation resistance	Major
c. Impact shock	Major
d. Transportation vibration	Major
e. High temperature	Major
f. Low temperature	Major
g. Humidity	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Continuity and resistance, and insulation resistance. Each test set shall be tested in accordance with 4.4.1 and 4.4.2. If any item fails to comply with 3.5 and 3.6, the item shall be rejected and removed from the lot.

4.3.3.2 Impact shock, transportation vibration, high temperature, low temperature, and humidity. One test set shall be selected from every lot of 40 or portion thereof, and shall be tested as specified (see 4.4.3). If any test set fails to comply with any requirement (see 3.8) the lot shall be rejected.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8877483). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

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a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Continuity and resistance.

4.4.1.1 Connector - J1. The circuit in the connector shall be tested by input voltage generated by the hand crank generator rotated to a speed not to exceed 5 revolutions per second. Using one #22 American wire gage size wire (length to suit), as a jumper, or with approved equipment connect the terminals as indicated in Table I - positions 2 through 18. Closed circuits - positions 2 through 18 - is indicated by illumination of the green "go" lens. Open circuit - position 1 is indicated by illumination of the red "no go" lens.

(Caution: Do not hold jumper, in place with hand, while testing).

4.4.1.2 Dummy connector recp (J2), Dummy connector (J3), Connector, Shorting (J4). Continuity between terminals listed in Tables II, III and IV shall be tested with an approved milliohmeter or equivalent, to determine compliance with 3.5.2, 3.5.3 and 3.5.4. Current shall be limited to 100 milliamps.

4.4.2 Insulation resistance. The insulation resistance shall be tested with an approved megohmeter at 500 volts direct current (vdc) + 50 vdc with current limited to 5 milliamperes (ma) maximum applied for a period of one minute maximum, 5 seconds minimum to determine compliance with 3.6.

4.4.3 Environmental testing (see 3.8).

4.4.3.1 Impact shock. The test set, enclosed in its carrying case, shall be securely mounted on a fixture capable of being readily positioned in each of the three mutuallly perpendicular axes. The test set shall then be mounted on a shock producing device and tested to the requirements in 3.8.1. Upon completion of this test, the test set shall be tested as specified in 4.4.1 and 4.4.2.

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4.4.3.2 Transportation vibration. The test set enclosed in its carrying case, shall be securely mounted on a fixture capable of being readily positioned in each of the three mutually perpendicular axes. The test set shall then be mounted on a vibration producing device and tested to the requirements in 3.8.2. Upon completion of this test, the test set shall be tested as specified in 4.4.1 and 4.4.2.

4.4.3.3 High temperature. The test set, in its carrying case, shall be placed into a temperature chamber. The temperature shall be raised to +160 ± 5°F and held at this temperature for four hours or until the temperature of the test set has stabilized (6.4). Remove test set from temperature chamber and test in accordance with 4.4.1 and 4.4.2.

4.4.3.4 Low temperature. The test set, in its carrying case, shall be placed into a temperature chamber. The temperature shall be lowered to -65 ± 5°F and held at this temperature for four hours or until the temperature of the test set has stabilized (see 6.4). Remove test set from the temperature chamber and test in accordance with 4.4.1 and 4.4.2.

4.4.3.5 Humidity. The test set with its case closed shall be placed into a humidity test chamber and exposed to humidity in accordance with MIL-E-5272, procedure I. The temperature specified shall be maintained within a tolerance of ± 5°F during the three humidity cycles. The humidity atmosphere shall be obtained by adjusting the pH value with chemically pure hydrochloric acid or chemically pure sodium hydroxide. Upon completion of the 3 cycles, the test set shall be removed from the chamber and excess moisture, wiped off. The test set shall then be tested in accordance with 4.4.1 and 4.4.2.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The test set shall be packaged in accordance with dwg 9221171 using the specified moisture barrier.

5.1.2 Level C. The test set shall be packaged in accordance with dwg 9221171 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The test set shall be packed in accordance with dwg 9221171.

5.3 Marking. Marking shall be as specified in accordance with dwg 9221171.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-2032 dated 7 May 1971, with Amendment 3 dated 26 June 1973.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1190-A099

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